Element C

Criteria for Variable Dimensions

Constraints

- The box should be 20" by 20" by 3' (to accommodate required facilities and mobility; listed as a constraint because other groups will expect a box this size to work with), and there should be 3 boxes total
- The box should hold a full-sized Komodo dragon without breaking
- The box should have guillotine-style doors (mandated by zookeepers)
- The box should allow the dragon to be seen in some way at all times
- The box must be finished by April 22nd

Criteria

Criteria	**Explanation**	**Method of Measurement**
Sturdy High (Highest)	The box should be able to hold the dragons without falling apart. Our group's task is primarily to develop the connection mechanism and gate architecture. This will be the weakest point of the box more likely than not—it shouldn't be the part that breaks. This is the most important because the box cannot succeed physically if it can't hold the dragon.	We could measure the max torque on the joint until it breaks. However, we might want to use a computer simulation to do this. We could stack weights up to 250 lbs in the box and test its ability to hold that load. Any sturdiness above this level is unnecessary, so we don't necessarily need to test it.
Adjustable High	The box should accommodate all dragons they intend to carry. This is second because the primary goal is restricted heavily if the solution isn't adjustable to all dragons.	Record the number of dragons the product could successfully accommodate out of all adult dragons. Accommodation indicates the box is adjustable to within 0.5 feet of the dragon's length (or allows the tail to stick out safely).
Light Medium	Two zookeepers will ideally be able to carry this box with the dragon inside, so the box can't weigh too much since the dragons already do. This is primarily important below the sturdiness because the primary	Weight (lbs)

	intent of the box is to easily and safely allow the transportation of the dragon. If too many zookeepers are required to pick up the box, the task will be difficult although possible to complete.	
Visibility Medium	Critical to the safety of the zookeeper: the dragon should be able to be seen at all times. If it begins to turn around (although it shouldn't be able to), the zookeeper should know. This is critical to the safety of the zookeepers but ranked below the other criteria because the zookeepers currently have no way of seeing the dragon in the tube they use, so the solution wouldn't change anything about their current method.	We could do a turbidity-style test. We could print numbers in varying fonts, asking people to read the smallest one they can and comparing it to the smallest font they could read without any material in the way
Cleanability Medium	The zookeepers indicated they would like a box that is easy to clean as they might need to disinfect the box after use. This goes here because it affects the long-term usability of the product. If the product cannot be cleaned, it will fulfill the primary goal, but it will not do so for a long time.	Composed of two parts: ● Bleach doesn't hurt the structural integrity of the product (zookeepers indicated they would likely use bleach)—add 1 cup bleach to the product, wait 2 minutes, wipe off, and retest "sturdiness" ● Bleach or other products will clean the product: Spill a cup of water colored with 5 drops of red food dye in the containment solution, wait 5 minutes, rerun the test of the bleach, and record the color of the material composing the box using some standardized color recording system (this is something that already exists)
Ease of Use	The box is more likely to be	Time to assemble completely

Medium	used as intended if it doesn’t require an immense amount of time to set up. The primary goal can be completed even if the box is hard to use, and it can be fulfilled indefinitely, but the solution will be less likely to be used. This means that this is the least important criterion.	(starting with all boxes separated and ending with the box fully assembled, one door open, and one door closed).
Aesthetics Low	The box might be slightly better if it appears better	We can ask 10 peers from different classes to objectively evaluate the box on a scale from 1 to 10 while defining 1 and 10 in terms of definite characteristics
Closable Distance Low	How far away the user can be and still close the door	We can attempt to close the door, and move back a foot upon success until we reach a point at which we cannot close the door. Asking 5 peers to do so (to account for differences in wingspan) and averaging the result should objectively test this.